Team 240

AI Incident Report Assistant: Final Report

Megathon X · Qualcomm Problem Statement 1 (On-Device AI Challenge) · Final submission

Summary. Sentinel AI is now a native Android, voice-first incident-reporting assistant demonstrated on the Snapdragon 8 Elite development kit. An employee describes an incident naturally; local models transcribe speech, identify one of 28 incident types, extract evidence-grounded facts, and ask targeted follow-ups. A validated report receives a responsible department, priority, service-level target and reference ID; the on-device facilities desk can review and update it. Since M2, the focus has moved from Kotlin logic validated with scripted model outputs to an integrated, measurable speech-to-report pipeline on Qualcomm hardware. No network is required at runtime. On-kit results include **1.78 s median speech-end-to-audible-reply latency**, **87.6%** incident classification accuracy, and **11 ms** NPU classification per turn. These results are from limited, mostly synthetic tests rather than a real-world field trial.

I. Final Solution and Evolution Since M2

The problem. Conventional workplace reporting forces a witness to pick the right department and complete a form while information is urgent and incomplete. A voice recording alone does not resolve missing safety facts or route the ticket. Sending internal speech to cloud services may also violate site privacy constraints.

The solution. Sentinel AI changes the flow to *Speak* → *Understand* → *Extract* → *Clarify* → *Review* → *File*. The reporter starts with one microphone control; live captions and a reactive aurora orb provide feedback. The assistant identifies the type, extracts only facts supported by speech, and asks for what is still needed. The employee can inspect/correct fields and confirm the local filing. Reports appear in an administrator/facilities queue sorted by priority, with status and response-due information. The role selector is a demo navigation aid, not production authentication.

What changed from M2. At mid-submission, the native Kotlin routing, review, persistence and admin state machine had passed **17 JVM tests and 57 emulator instrumentation tests**, but live ASR, classification, extraction and TTS had not been exercised end-to-end. In M3, the demonstration uses an Android/Kotlin app with on-device AI components connected through native bridges. The former Python prototype remains development/test scaffolding; there is no Python server in the runtime path. A real device demonstration and stage-by-stage measurements now accompany the earlier application tests.

II. Technical Approach

1. Capture and turn-taking. Audio is captured locally with noise suppression and echo cancellation. A pause detector (nominally 0.9 s) closes a conversational turn; responses are spoken, and the microphone can reopen for the next answer. The UI supports interruption and shows the current transcription/form state.

2. Speech recognition. Whisper base performs ASR with its encoder on the Hexagon NPU and decoder on the CPU using `whisper.cpp`. Captions provide visible feedback. The current tested release is English-only; Hindi/Hinglish remains a next step.

3. Fast incident understanding. A fine-tuned Laya classifier (ModernBERT-large with LoRA, W8A16 on NPU) predicts the incident type, relevant sentences and field mentions. Confidence-based clarification avoids silently selecting an uncertain category. The taxonomy has **28 incident types across six responsible departments**; irrelevant requests are declined rather than filed.

4. Evidence-grounded extraction. Qwen3-0.6B Q8_0 runs through `llama.cpp` on the CPU and proposes grammar-constrained structured values. Validation retains values only when supported by transcript evidence and legal schema values. Absent details remain missing instead of being hallucinated. The classifier identifies the type; deterministic taxonomy rules decide routing, escalations, notifications and SLA.

5. Controlled dialogue and filing. The policy asks for missing essentials, prioritizes safety details when applicable, limits repeated questioning, and accepts explicit unknown answers. Piper TTS via `sherpa-onnx` speaks the next question. After review/confirmation, the local filing function generates a reference ID and stores a report with status and respond-by time. This is local workflow state, *not* cross-device dispatch or cloud delivery.

III. Key Product Features

- **Hands-free reporter flow:** one-tap voice reporting, automatic conversational follow-ups, live captions and an audio-reactive aurora visualizer.
- **Live structured form:** WHAT/WHO/WHERE progress, auto-filled fields with evidence, missing/unknown states and tap-to-correct validation.
- **Configurable response logic:** 28 form types, safety-sensitive priority P1–P4, six owning teams, assignee roles and response SLAs defined by local taxonomy JSON.
- **Offline facilities desk:** department/status filters, incident details, read-aloud, status updates and locally stored history.
- **Privacy by design:** inference and filing stay on the Snapdragon device; no runtime network calls and no transcript logging to system logs.

Location validation. The Qualcomm-HQ demo form pack permits only **Building N (floors 1–12)** and **Building QRC (floors 1–6)**. The assistant checks building and floor and obtains a specific room/area identifier; invalid or misheard locations are clarified, never guessed. These building/floor values are *team-defined demo configuration*, not verified Qualcomm facility data.

IV. Incident Workflow and Safeguards

Worked example. A reporter says, "There is water dripping from the ceiling near the pantry, in N building on the fourth floor." The pipeline can infer a water-leak incident, Building N, Floor 4, the pantry as a location cue, and the ceiling as leak source. It must *not* invent a burst pipe or nearby socket. Before filing, the assistant asks whether the water is spreading, whether electrical equipment is nearby, and which precise room or area is affected. If proximity to electrics is confirmed, the taxonomy escalates the ticket from its default P3 level to P1 and adds electrical and EHS notification roles. This example illustrates the difference between model-generated suggestions and validated report fields.

Fixed location scope. The two allowed demo codes are N (1–12) and QRC (1–6); building, floor and specific space are essential. "N" and "QRC" are recognized via priming/normalization, but confirmation is preferable when ASR confidence is low. The assistant must explicitly seek corrections for invalid floors, rather than substituting another building. The pack is a configurable campus demonstration, not a source of actual Qualcomm site maps.

Safety before convenience. Incident templates define required safety fields: e.g. a water leak asks about spread and electrical proximity; an elevator fault asks whether people are trapped; a fire report asks about visible flames and evacuation. The dialogue asks safety-critical facts before other missing details, limits follow-up turns to four and question width to two fields, and permits "I don't know" without silently fabricating a safe answer. Unknown safety facts trigger conservative priority handling.

On-device function contract. The classifier supplies an incident-type candidate; Qwen proposes legal typed fields with source spans; deterministic Kotlin logic validates inputs and calculates owner, priority, response deadline and co-notification roles. The UI displays the draft and allows corrections. Only after explicit confirmation does the local filing operation assign a reference ID and put the incident into the on-device desk queue; this does not imply that a remote responder was alerted.

V. Results and Verification

Table 1: Measured on Snapdragon 8 Elite kit (M3).

Metric	Observed result
Speech end to reply audible	1.78 s median; 2.79 s p90
Turn detected to reply audible	1.0 s median
Incident type accuracy	87.6% (held-out synthetic set)
Classifier on Hexagon NPU	11 ms/turn; 47× faster than CPU
ASR word-perfect turns	19/23 (83%); mean WER 2.8%
Model readiness after launch	Approx. 1.4 s
Resident model memory	Approx. 2.5 GB
Voice-filed conversations	5/6; one stalled on "QRC"

Stage-level latency. Median reported components were approximately 786 ms end-of-turn pause detection, 191 ms Whisper encoder, 11 ms classification, 906 ms Qwen extraction, 409 ms question writing and 57 ms to first TTS audio. These per-stage medians are not additive to the end-to-end median because measurements were captured at different boundaries and across different turns. A 1.0 s measured median from turn detection to reply audible reflects the integrated pipeline rather than a sum of independent column medians.

Benchmark method. Latency was sampled over **23 turns from six scripted conversations**, with clean synthetic audio injected into the actual voice engine at real-time pace. The pause gate accounts for approximately 0.79 s of total median response time. The first conversational turn is slower (2.73 s) than later follow-ups (1.64 s) because extraction processes more accumulated context. The classifier's held-out evaluation also uses LLM-written conversations; figures should not be interpreted as real-workplace accuracy.

M2-to-M3 verification trail. The M2 architecture had already replaced a Python-only prototype with a native Kotlin application. Its routing/review/admin behavior had 17 JVM and 57 emulator instrumented tests with no failures, but real speech and device latency were untested. At M3, on-kit tests added ASR/voice loop measurements, classifier timings and a photographed hardware demonstration. The earlier passing logic tests and new kit benchmark measure different things; neither alone establishes robust performance in production workplaces.

Hardware proof. The model stack was demonstrated on the Snapdragon 8 Elite (SM8750P) development kit using the Hexagon NPU for Whisper's encoder and Laya; Qwen and Piper execute on the CPU. The Android app is Kotlin/Jetpack Compose, with JNI integration and local report storage. Figure 1 shows the voice reporting interface on the attached development kit.

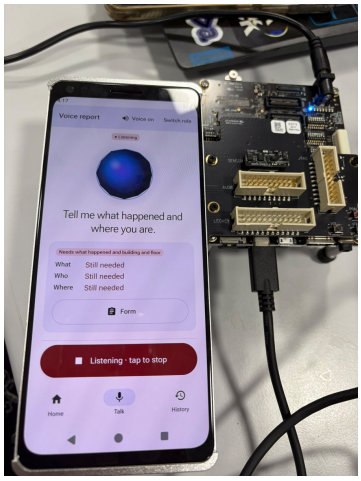


Figure 1: Sentinel AI voice-reporting prototype running beside the Snapdragon development kit (team photo).

VI. User Interface and Demo Evidence

Reporter view. A role-selection landing page opens Employee mode. The recording screen provides mic feedback, a voice-reactive aurora orb, partial caption and a progressively filled WHAT/WHO/WHERE form. Missing-field prompts are spoken aloud. The confirmation screen lets the user inspect and correct a specific value before the saved ticket is created. A visible role selector is for demonstration purposes and is not a security or authentication boundary.

Facilities view. The administrator/facilities desk presents open work in urgency order, with filters, incident location, owning team, deadline and status. A responder can open a report and move it through the local work-state flow. Administrative data represents the same on-device store as reporter history; live communication to another device or server is deliberately excluded from this release. Together with Figure 1, Figure 2 documents both ends of the interaction rather than a standalone recording mockup.

VII. Limitations, Next Steps and Final Deliverable

Current limitations. The demo is English-only. Single-letter building codes are vulnerable to ASR errors ("QRC" caused one of six benchmark scenarios to stall); nearby speech can be mistaken for a turn. Extraction on the small general-purpose Qwen model can miss fields. The six-conversation timing sample and synthetic test corpus are limited; independent human-spoken testing is necessary. Server/ticketing-system sync, production authentication and indoor positioning are not part of v0.1.

Evaluation caveats. Five of the six scripted conversations com-

pleted filing by voice, but the failure on "QRC" demonstrates a meaningful error mode rather than a cosmetic issue. Only 23 clean, synthetic voice turns informed the ASR and latency report; real speech, accents, workplace noise and bystanders require independent evaluation. The classifier evaluation set is also synthetic and written by LLMs. The current 87.6% result therefore describes held-out in-domain data, not a deployment-grade field accuracy estimate.

Scope boundaries. No production account authorization, inter-device synchronization, real-time responder paging, or indoor location tracking has been demonstrated as a released feature. The campus taxonomy, departments and SLAs are locally configured example policy values. The main safety contribution is constrained reporting and conservative escalation, not an autonomous emergency response capability.

Next steps. Improve location recognition with domain priming and confirmation; evaluate real voices/noisy rooms; fine-tune extraction and reduce latency; add Hindi/Hinglish, optional server sync and per-site taxonomies. On-device privacy remains a fixed requirement.

VIII. Operational Behavior and Integrity

Classification and ambiguity. The incident taxonomy gives the classifier a closed world of 28 valid types plus outcomes for undetermined and out-of-scope reports. An uncertain classification leads to a spoken choice among likely categories instead of an unsupported assignment. Multiple incidents in the same narration require a separate signal that the speaker actually described distinct events; merely naming two plausible labels is not sufficient. This distinction mattered during M2 testing because an ambiguous single report initially created an unwanted second draft. The corrected workflow separates ambiguity from multiplicity.

Correction and provenance. A voice transcript is evidence, not the final report by itself. Each extracted field is associated with the passage that supports it; otherwise the value remains unfilled. When a reporter corrects an attribute, any derived values such as priority or dispatch role must be recalculated against the current form and taxonomy. The review stage avoids saving stale priority decisions made from earlier, incorrect facts. A local draft is distinct from a filed report; the submission action must be idempotent to avoid duplicate incidents if the user presses the control twice or a callback repeats.

Urgency calculation. Priorities P1–P4 correspond to nominal response targets of 15, 60, 240 and 1,440 minutes in the team-defined pack. The default comes from the incident type; deterministic rules may raise it when a safety flag is true, with the most urgent matching rule taking precedence. For example, an electrical hazard defaults to P1, whereas a water leak normally starts at P3 but can escalate when cables or sockets are threatened. The incident record carries department, responsible role, relevant secondary notifications and response-due timestamp calculated from the local device time.

Work queue consistency. The reporter's history and the facilities desk are two views over local incident records, not an internet-delivered workflow. Desk staff can inspect the structured report and update its work status, which is then reflected in the local history. Presentation components remain separated from the field-extraction model so that a future site can replace the on-device taxonomy and deploy its own form types without redesigning every view. Statuses describe internal workflow progress; they should not be described as proof that outside emergency services received any message.

IX. Testing Interpretation and Practical Risks

What the measurements establish. The Snapdragon evaluation confirms that the described models can execute within the available device resources and that a spoken end-to-end conversation can complete on the kit. The 11 ms classifier inference time is a component measurement, whereas the 1.78 s median audible reply includes conversational turn detection and downstream stages. Five of six benchmarked scripted conversations reached local filing. These are useful engineering milestones, not guarantees about unseen incidents or all workplace acoustic conditions.

What remains unproven. The classifier's synthetic test language is cleaner than spontaneous employee speech. ASR was measured on 23 clean turns, and confidence may change with accents, code-switching, alarms or other talkers. Single-letter building codes are especially brittle. The product should repeat uncertain locations, favor explicit confirmation of safety facts and allow users to revise their report. Expanded testing should include fresh human-spoken cases, noisy locations and operational review of misroutes before workplace deployment.

Near-term engineering plan. Prioritize accurate recognition of N/QRC and specific room identifiers; assess real-voice end-of-turn detection; improve Qwen field completeness without relaxing evidence checks; and verify interruption, repeated submission and longer sessions under memory/thermal pressure. Planned Hindi/Hinglish recognition, per-site form packs and server synchronization are future extensions, not completed M3 deliverables.

Engineering handover. Reproduction requires the Kotlin/Compose repository, JNI bridges for whisper.cpp/llama.cpp, the Hexagon deployment artifacts and supplied QAIRT/SNPE runtime, the local 28-type taxonomy, and the documented model bundle (approximately 2.2 GB). The kit uses Snapdragon 8 Elite-class hardware; measured resident footprint was approximately 2.5 GB. Device- and runtime-specific proprietary binaries are supplied through the appropriate build environment rather than embedded as unrestricted public assets.

Final handover. One native Android app and source repository; build/run instructions and runtime/model setup; taxonomy and deployment configuration; a device demo and backup recording; plus documented latency, classification and ASR results. The functional achievement is a **fully local voice-to-structured-report loop**, from initial speech to reviewed, classified and priority-routed incident history.

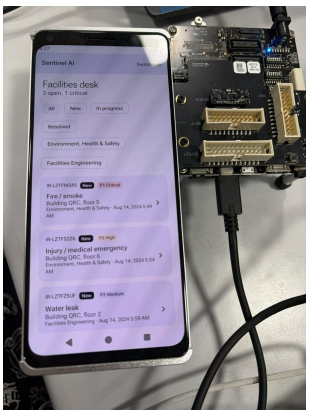


Figure 2: Local facilities-desk queue with team, severity and incident status filters (team photo).